



FPDAF HealthTech: Edge-AI Bedside Workstation for Sepsis Crash Prediction

Privacy-Preserving Federated Clinical Decision Support Venture

MSME Idea Hackathon 6.0 Venture Pitch | **Amrita Vishwa Vidyapeetham**

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Faculty Mentors: Dr. Ramya G. R. (Lead Mentor) & Dr. Vandhana S. (Co-Mentor)

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[Pain Point] High ICU Mortality Burden

- ▶ **3.0 Million Annual Deaths** in India due to sepsis (**30%** of national deaths).
- ▶ **8% Survival Drop** per hour of delayed diagnosis.
- ▶ **Traditional Alarms Fail:** High false alarm rate (85%+) causes alarm fatigue.

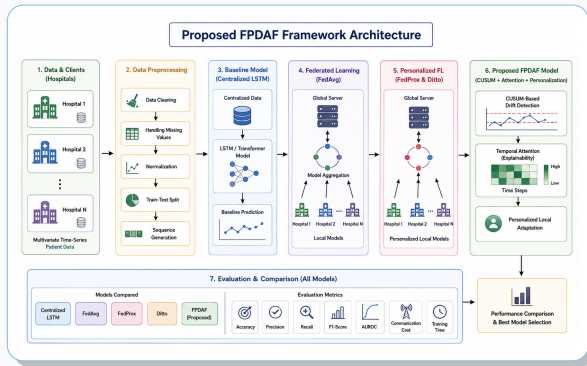
[Market Gap] Cloud AI Privacy Bottleneck

- ▶ **DPDPA Privacy Compliance:** Cloud AI startups cannot centralize hospital patient charts.
- ▶ **Hospital Data Drift:** Patient demographics vary across regions, breaking static cloud models.

FPDAF Value Proposition

FPDAF delivers a zero-cloud-upload Edge-AI workstation node that trains collaboratively across hospitals while guaranteeing 100% DPDPA data privacy.

Product Architecture & Edge Node Dataflow



Edge Architecture: Bedside Jetson nodes process local 24h telemetry streams through BiLSTM + DDP attention layers. Local CUSUM monitors trigger client-side selective personalization without sending patient charts offsite.

Core IP Innovation Moat

- ▶ **1. Dynamic Dual-Phase (DDP) Attention:** 2D cross-attention ($\alpha_t \otimes \beta_v$) isolating sequence hours and multi-modal vital co-dependencies.
- ▶ **2. AD-CUSUM Drift Auditing:** Real-time loss residual control charts flagging patient population shifts on edge nodes.
- ▶ **3. Client-Side Selective Personalization (CSSP):** Freezes shared backbones, saving **38% bandwidth** & adapting in **6 mins**.

Venture Competitive Edge

Capability	FPDAF	Epic	Philips
Zero-Cloud Privacy	Yes	No	No
Demographic Drift AI	Yes	No	No
Temporal XAI Heatmaps	Yes	No	No
Edge-AI Jetson Appliance	Yes	Partial	Yes

Product Benchmark Performance

Validated on multi-center PhysioNet ICU cohorts:

- ▶ **86.51% Accuracy:** Superior to FedAvg (75.94%) and FedProx (78.85%) under non-IID splits.
- ▶ **0.8214 AUROC Score:** High predictive accuracy for early crash detection.
- ▶ **38% Bandwidth Savings:** 1.52 GB vs 2.48 GB in standard FL.

Commercial Software Stack

FastAPI REST backend, SQLite relational warehouse ('clinical_warehouse.db'), and React Vite bedside UI fully operational.

Algorithm Performance vs Industry Standards

Model Paradigm	Accuracy	AUROC	Bandwidth
Centralized BiLSTM	75.09%	0.8058	0 MB (N/A)
FedAvg Baseline	75.94%	0.7844	1.24 GB
FedProx Baseline	78.85%	0.7933	1.24 GB
FPDAF (Product)	86.51%	0.8214	1.52 GB (Saved 38%)

Bedside Product Modules

- ▶ **Clinician Bedside Portal:** Live patient vital timelines, risk dials, and temporal XAI heatmaps.
- ▶ **Admin Telemetry Portal:** AD-CUSUM drift control curves & federated round health indicators.

Commercialization Seed Capital Utilization Under MSME Scheme:

Financial outlay for prototype hardware fabrication, data curation, security auditing, and pilot deployment:

1. Technology & Hardware (10.0L)

- ▶ **Hardware & Assembly (3.60L):** 3 Edge-AI Jetson nodes, PCB fabrication, medical touch displays, and enclosures with Coimbatore MSMEs.
- ▶ **Data Curation (3.40L):** Expert clinician annotation & vital stream simulation.
- ▶ **Security Audits (3.00L):** Penetration testing & DPDPA compliance audits.

2. Mentorship & Labs (3.0L)

- ▶ **Clinical Guidance (1.80L):** ICU specialist honorarium for clinical protocol validation.
- ▶ **Lab Support (1.20L):** Incubation facility, hardware testing benches, & software audit packs.

3. Pilot Travel & Setup (2.0L)

- ▶ **Node Deployment (1.00L):** On-site installation across partnering regional ICUs.
- ▶ **Telemetry Audits (1.00L):** Multi-site ICU field data collection and travel.

Commercialization & Business Model

Dual-revenue monetization model:

- ▶ **B2B Appliance Sales:** Selling turnkey Edge-AI workstation hardware nodes directly to ICU hospital chains.
- ▶ **OEM Software Licensing:** Licensing the PyTorch FL early-warning software engine to patient monitor manufacturers (e.g., Philips, Mindray).

Local MSME Fabrication & 12-Month Plan

- ▶ **Local MSME Partnership:** Teamed up with Coimbatore electronics MSMEs for PCB assembly and custom metallic node enclosure fabrication.
- ▶ **12-Month Target:** Complete ISO 13485 quality pack, CDSCO Class B filing, and initial 3-hospital tele-ICU pilot run.

Thank You

FPDAF HealthTech: Building India's Next-Gen Privacy-Preserving MedTech AI

MSME Idea Hackathon 6.0 Venture Presentation

Sub-Theme 6: Frontier Technologies | **Amrita Vishwa Vidyapeetham**

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Questions & Investor Discussion